Microservices Project - Part 2

The second phase of this project will address the internal aspects of connectivity, security, observability and deployment.

To achieve this you will deploy a service mesh into your project and configure it accordingly.

Kubernetes

The first stage is to deploy your project in Kubernetes. You can deploy to a local environment such as KinD, such that you can demonstrate the application running in a cluster.

Use stateless and stateful workloads as appropriate, with an ingress gateway for external connectivity.

Ensure your application can be started with a tool like Kustomization or Helm so the application can easily be started in KinD or Minkube for assessment.

Istio

Once your project is deployed in Kubernetes, you will meet the objectives of the project by deploy the service mesh Istio.

Your service mesh should address the following:

- Secure communications:
    - mTLS on inter service communications
    - Zero trust through authorisation (traffic only comes from intended workloads)
- Traffic Control:
    - Load balancing
    - Canary deployment of new releases
- Observability:
    - Setup key metrics
    - show distributed tracing
    - Visualise cluster topology
- Implement Resilience Features:
    - Rate limiting
    - Circuit breaker
- Fault/Chaos Testing:

- Evaluate impact of system degradation (e.g. latency)

Report

Document your implementation, with test cases to demonstrate that all features above have been implemented.

Your report should give an overview of Istio's architecture and how it provides the capabilities demonstrated in your application.

Include description of how security model is working and how Istio (with Envoy) controls and directs traffic within the cluster.

Describe and compare the Side Car vs the Ambient implementation of Istio, with pros and cons of each. Describe L4 vs L7 and how L7 can be address in ambient configuration.

Provide a Git repo to your final project and a Word or pdf document for your report. Do not use AI for generating the report.